

DBMS Lab 0

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Q 1 What is SQL and its Use?

SQL is a standard language for accessing and manipulating databases.

SQL stands for Structured Query Language, which lets you access and manipulate data. It can perform CRUD operations i.e. Create, Read, Update and Delete the Data.

One can create multiple tables with different data types in separate rows and columns. It can create various views in the database and can set permissions on the tables, procedures and views.

RDBMS (Relational Data Base Management System) is the basis of SQL and for all other modern database systems such as MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.

Q 2 Describe all the different commands of SQL, mention use of it and syntax.

SQL allows various actions which are acted upon via the use of SQL statements. SQL statements contain basic keywords and end with a semi colon “;”.

Here are some examples of the keywords used in SQL :-

- **SELECT** - extracts data from a database

```
SELECT CustomerName, City FROM Customers;
```

- **UPDATE** - updates data in a database

```
UPDATE table_name SET column1 = value1, column2 = value2  
WHERE condition;
```

- **DELETE** - deletes data from a database

```
DELETE FROM table_name WHERE condition;
```

- **INSERT INTO** - inserts new data into a database

```
INSERT INTO table_name VALUES (value1, value2, value3,  
...);
```

- **CREATE DATABASE** - creates a new database

```
CREATE DATABASE database_name;
```

- **ALTER DATABASE** - modifies a database

```
ALTER DATABASE OldDatabaseName MODIFY NAME =  
NewDatabaseName;
```

- **CREATE TABLE** - creates a new table

```
CREATE TABLE table_name (column1 datatype constraint,  
column2 datatype constraint, column3 datatype constraint);
```

- **ALTER TABLE** - modifies a table

```
ALTER TABLE table_name ADD column_name datatype;
```

- **DROP TABLE** - deletes a table

```
DROP TABLE table_name;
```

- **CREATE INDEX** - creates an index (search key)

```
CREATE INDEX index_name ON table_name (column1, column2, ...);
```

- **DROP INDEX** - deletes an index

```
DROP INDEX table_name.index_name;
```

Q 3 Write a step by step procedure to install MYSQL.

1. Download the MySQL Installer from the web..
2. Run the installer and select the required setup type from the download window.
3. Install the required MySQL components.
4. Configure MySQL Server.
5. Set a password for the root user.
6. Complete the configuration process.
7. Open MySQL Workbench.
8. Connect using the MySQL credentials.
9. Verify installation using: SHOW DATABASES;

If databases are displayed, MySQL installation is successful.